Causal effects of schools on higher education outcomes - Spring update

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1 Introduction

2 Change of Approach: from school-level effects to 1 dimension.

3 Methodology

3.1 Computing Observational Value-Added for each school

3.2 IV Regression: Projecting lottery induced variation into Value-Added

3.3 Characterization of Compliers

4 Preliminary Results

4.1 Core Results

4.2 Characterization of Compliers

5 Extensions

5.1 Are individuals induced to STEM fields by their schools better off?

5.2 Implement Empirical Bayes

6 Conceptual Framework and Identification Strategy

In this section I describe how I am currently thinking of my analysis. I review the assumptions and methodologies related to the estimation of school specific effects using centralized assignment mechanisms, and then I present the main reduced-form regressions I intend to run, as well as some challenges and how I intend to address them.

6.1 Potential Outcomes and School Effects

For this section I borrow from **angrist_credible_2024** and **abdulkadiroglu_research_2017**

Suppose N students and J schools. Following the value-added literature on school effectiveness, I assume a constant-effects model. Then, the potential outcome Y_{ij} of a student i in a school j is additively separable and can be written as:

$$Y_{ij} = \nu_j + \epsilon_i \tag{1}$$

Hence, the causal effect of attending a school j relative to k would be given by $\nu_j - \nu_k$.

By adding a constant, we can renormalize this model relative to the average achievement ($\beta_0 = \frac{1}{J}\sum_{j=1}^{J}\nu_j$) As a result, we can rewrite the model as:

$$Y_i = \beta_0 + \sum_{j=1}^{J}\beta_j D_{ij} + \epsilon_i \tag{2}$$

where now $\beta_j = \nu_j - \beta_0$.

Unless school assignment is completely random, the evident threat to running this regression is the fact that $\mathbb{C}(\epsilon_i, D_{ij})$ might be different than zero. This selection bias would bias our estimates and prevent a causal interpretation.

6.1.1 Assignment Mechanisms and Assignment Risk

A common strategy used in research to overcome this issue is the use of individual school lotteries, and more recently, centralized school assignments. In centralized assignment mechanisms, applicants submit rank-ordered preferences over schools ($\succ_i$) at which they might also have different priority levels[1]. These priorities operate as the school's preferences for students. An algorithm in place (most commonly student-proposing Deferred Acceptance) takes these preferences and outputs a school assignment (Z_{ij}) with one offer per student. In order to break ties of students with the same preference and priority level, these systems randomize.

If we refer to a student's priority at each school as $\rho_i = (\rho_{i1}, ..., \rho_{iJ})'$, we can define a student's type as the combination of both $\theta_i \equiv (\succ_i, \rho_i)$. Let the probabilities of admission at a school be given by $p_i = (p_{i1}, ..., p_{ij})'$, with $p_{ij} = Pr(Z_{ij} = 1)$.

The mechanism described above implies that students with the same preferences and priority vector have the same probability of being assigned to the schools in their portfolio, which is called by **abdulkadiroglu_research_2017** the 'equal treatment of equals' (ETE) property. Formally ETE states $\theta_i = \theta_j \implies p_i = p_j$.

They also prove that under ETE that for any observed characteristic fixed under re-randomization W_i (i.e. pre-assignment or not affected by assignment) we have that:

$$Pr[Z_{ij} = 1 | \, W_i = w, \theta_i = \theta] = Pr[Z_{ij} = 1 | \, \theta_i = \theta] \tag{3}$$

Hence, for any such variable W_i we can write

$$W_i \perp\!\!\!\perp Z_i \,|\, \theta_i \tag{4}$$

In particular, and as a consequence of equation 1, where we assumed ability ϵ_i to be fixed under re-randomization, partly due to the absence of match effects, we can write

$$\epsilon_i \perp\!\!\!\perp Z_i \,|\, \theta_i \tag{5}$$

[1] e.g. in the Chilean case being the child of a member of the school staff gives a student priority in that school.

This is the key identifying assumption of the design. It states that conditional on the student type, the school offer is exogenous to the error term (in this example student's ability). Under this assumption and perfect compliance ($Z_i = D_i$), we could estimate equation 2 by saturating on an indicator for each student type θ_i.

However, that is often not feasible as these strata are too granular to have enough variation within each of them to estimate the model. As described by **angrist_credible_2024**: "In practice (...) we see nearly as many preference and priority combinations as there are students in a match." A solution to this issue is to use the probabilities of admissions as propensity scores to pool over student types, as proposed by **abdulkadiroglu_research_2017**. This is an application of the propensity score methods (**rosenbaum_central_1983**) and it yields:

$$\epsilon_i \perp\!\!\!\perp Z_i \,|\, p_i \tag{6}$$

Note that by eq. 4 we can also consider other control variables (as long as they are fixed under re-randomization). Thus for any pre-treatment covariate X_i it must be the case that $(\epsilon_i, X_i) \perp\!\!\!\perp Z_i|\theta_i$, so we can rewrite Eq 6 as $\epsilon_i \perp\!\!\!\perp Z_i \,|\, (p_i, X_i)$ to allow for such controls.

6.1.2 Towards Conditional Independence Assignment

Although an instrumental variables approach may seem natural given the availability of many exogenous instruments, estimation of school-specific effects suggests otherwise. The issue arises from the multiplicity of (unordered) outside options. Individuals instrumented into a school may come from different margins, and as a result interpretation becomes challenging unless strong assumptions are made (**angrist_methods_2023**) (**kirkeboen_field_2016**). To see this, consider students with an identical ranked list of three schools who are guaranteed admission to their third choice if they are not successful in their first two applications. An offer to the first school then generates two margins of compliance. The problem becomes more complex when students have different fallback options or when identical preference lists are rare in the data.

Instead, to compute effects at the school level, the literature typically relies on selection on observables i.e. the construction of a control vector X_i such that $\epsilon_i \perp\!\!\!\perp D_i \,|\, (X_i)$. Note that under the constant effects assumption and the ETE property of lotteries, equation 6 is closely related to this framework.

An additional assumption that allows us to move from equation 6 to a conditional independence framework is that compliance with offers is as good as random. The argument for this assumption is that all sorting occurs at the application stage, and thus captured by the individual type θ_i, and that post-offer compliance is determined by idiosyncratic shocks uncorrelated with the error term (in this case ability ϵ_i). This argument is often credited to **dale_estimating_2002**.

$$\epsilon_i \perp\!\!\!\perp D_i \mid (p_i) \tag{7}$$

Compliance being as good as random yields a traditional conditional independence assumption, from where estimation is straightforward.

6.2 Estimation in specific outcomes

In this subsection I present the main regressions I intend to run, as well as an approach to provide reduced-form evidence for a non-score mechanism of schools on major choice. Let Y_i^{MATH} represent the outcome of a student in the standardized exam for mathematics and Y_i^{STEM} an indicator variable for whether the student enrolls in a STEM program.

Let X_i be a vector of characteristics not changed by treatment (e.g. demographics), and $Y_{pre,i}^{MATH}$ a measure of pre-treatment math achievement .

Following equation 7 I write:

$$Y_i^{MATH} = b_0 + \sum_j b_j D_{ij} + X_i' B + b_{pre} Y_{pre,i}^{MATH} + g(p_i) + \epsilon_i \tag{8}$$

$$Y_i^{STEM} = c_0 + \sum_j c_j D_{ij} + X_i' C + g(p_i) + \eta_i \tag{9}$$

The only difference between these two regressions is the fact that I propose to include a lagged outcome (pre-treatment math achievement) on the first regression. I do not have a similar lagged outcome for STEM enrollment. The inclusion of lagged outcome for math achievement makes 8 an 'RC VAM' regression under the **angrist_credible_2024** terminology. While 9 is a Risk-Controlled OLS or what **angrist_credible_2024** calls 'Risk only'.

Remember that, under the CIA assumption above, these regressions could be run without any of the controls and should still hold causal interpretation: the CIA assumption only needs the p_i to be included.

6.2.1 Thought exercise: what happens if math achievement affects major choice?

Suppose that in the true model for STEM enrollment we have the student outcome in the standardized math exam Y_i^{MATH} entering linearly, and that everything else is specified as in equation 9. In such case, the true model would be given by:

$$Y_i^{STEM} = \gamma_0 + \sum_j \gamma_j D_{ij} + \gamma_M Y_i^{MATH} + X_i' \Gamma + g(p_i) + \eta_i \tag{10}$$

This idea is supposed to capture relationships of sorting based on ability, as well as eligibility cutoffs.

Then, if we run the regression 9 we would get that:

$$\mathbb{C}(Y_i^{STEM}; D_{ik}|X_i, p_i) = \mathbb{C}(\gamma_0 + \sum_j \gamma_j D_{ij} + \gamma_M Y_i^{MATH} + X_i'\Gamma + g(p_i) + \eta_i; D_{ik}|X_i, p_i)$$

$$= \mathbb{C}(\gamma_0 + \sum_j \gamma_j D_{ij} + \gamma_M \Big[b_0 + \sum_j b_j D_{ij} + X_i'B + g(p_i) + \epsilon_i\Big] + X_i'\Gamma + g(p_i) + \eta_i; D_{ik}|X_i, p_i)$$

$$= \mathbb{C}(\gamma_k D_{ik} + \gamma_M \Big[b_k D_{ik} + \epsilon_i\Big] + \eta_i, D_i j|X_i, p_i)$$

$$= \mathbb{C}(\gamma_k D_{ik} + \gamma_M \Big[b_k D_{ik}\Big] +; D_{ik}|X_i, p_i)$$

$$= \Big(\gamma_k + \gamma_M(b_k)\Big)\Big(\mathbb{V}(D_{ik}|X_i, p_i)\Big)$$

$$\neq \gamma_k \Big(\mathbb{V}(D_{ik}|X_i, p_i)\Big)$$

Where the first equality follows from replacing equation 8 in 10. The second one from dropping constants and functions of the conditioning set. The third one from applying equation 7 to the individual's fixed mathematical ability (ϵ_i) and interest for STEM (η_i), and the final one from covariance/variance identities.

Note that unless either $\gamma_M = 0$ (math scores not affecting likelihood of student applying to STEM) or $(b_k) = 0$ (school not affecting math scores), the outcome of the short regression would include both an effect through scores, as well as a potential non-score effect.

If we run regression 9, however, for a subset of schools for which we have estimated $b_j = 0$, we are effectively shutting-down that channel.

Thus, any non-zero c_j estimated for those schools would provide evidence of a non-score effect of schools on major choice.

7 A Structural Model of Major Choice

An alternative approach to separate these two channels is to estimate a model of demand for fields of higher educational choices, following **abdulkadiroglu_parents_2020** and **campos_college_2026**. Note that since I am trying to study major choice as an outcome, I am moving away from lists for high school admission into ranked-lists of preferences for higher educational programs. I, thus, follow most closely the work by **campos_college_2026** as I decide to collapse the demand into broad fields of occupation as opposed to particular programs. Moreover, the authors use data from the same context: ranked-lists for higher education programs in Chile.

Let u_{if} be the indirect utility associated with field $f \in \mathcal{F}$. Let $g(i)$ index the covariate-cell group of i. I specify the following functional form:

$$u_{if} = \delta_{g(i)f} - \kappa_{g(i)} d_{if} + f_{g(i)f}(Y^{MATH}, Y^{VERBAL}) + \lambda_{fs} + \eta_{if} \tag{11}$$

where $\delta_{g(i)f}$ represents the mean utility of field j for applicants of covariate-cell $g(i)$, d_{if} is given by the distance between student i and the nearest program corresponding to the field f, and $\kappa_{g(i)}$ is a cell-specific distance cost. $f_{g(i)f}(Y^{MATH}, Y^{VERBAL})$ gives a

polynomial of degree 2 for the math and verbal exams, for each covariate-cell and field. λ_{fs} is a school-field preference shifter. Finally η_{if} represents the unobserved preference shock, assumed to follow an EVT1 distribution conditional on (X_i', d_i'). The first two terms $(\delta_{g(i)f}, \kappa_{g(i)})$ are relatively standard in these exercises and included on both models cited above. The polynomial of degree 2 for each exam, field studied and covariate cell allow to flexibly account for how scores might play an important role in the choice of majors. Scores affect elegibility for particular programs, measure skills that might be relevant to them, while also potentially operating as a signal for students of their comparative advantage. Finally the λ_{fs} is the main object of interest in this exercise, as it allows to test whether schools do shift their students preference even when allowing scores to have a very flexible effect.

The rationale underlying this approach is that submitted college application ranked-choice lists as revealing their preferences given that in this setting truthful reporting is a weakly dominant strategy. This is given by the fact that the mechanism used to process them is a student-proposing DA system. If we refer to a student's rank-order list by $R_i = (R_{i1}, ..., R_{il(i)})'$ where $l(i)$ is the number of different fields ranked by the student, then we have that the students' top-ranked field satisfies we have that:

$$R_{1i} = \arg\max_{f \in \mathcal{F}} U_{if}$$

with the remaining options satisfying

$$R_{ki} = \arg \max_{f \in \mathcal{F} \setminus (R_{i1}, ..., R_{k-1})} U_{if}$$

The EVT1 assumption implies the following conditional likelihood of the rank list R_i:

$$\mathcal{L}(R_i | X_i, d_i) = \prod_{k=1}^{l(i)} \frac{\delta_{g(i)f} - \kappa_{g(i)} d_{if} + f_{g(i)f}(Y^{MATH}, Y^{VERBAL}) + \lambda_{fs}}{\sum_{r \in R_i \setminus R_{i1}, ..., R_{k-1}} \delta_{g(i)f} - \kappa_{g(i)} d_{if} + f_{g(i)f}(Y^{MATH}, Y^{VERBAL}) + \lambda_{fs}} \tag{12}$$

For the covariate-cells, I consider each combination of macro-region (3 values), gender (2 values) and family income (above, below median: 2 values) for a total of 12 covariate-cells. For each of them, I need to compute 5 parameters for each field -mean utility and 2 polynomials of degree 2- plus an extra term for the sensitivity to distance. That would yield a total of 12 x (5(8) + 1) = 12(41) = 492 parameters. On top of that estimating school-preference shifter with 200 schools, would require other (200)(8) = 1600 parameters, for a total of 2092 parameters. Limiting myself to study only 100 schools would bring down the number of parameters to 1292.

An important caveat is that this approach does not rely at all in the use of lotteries. As a result, as long as preferences are relatively stable over time, I could use data from cohorts that arrived to high school before the introduction of the lotteries. Potentially, this is also an interesting space to work on.

8 Computing Assignment Probabilities

Evidently, as discussed in section 6, the computation of the p_{ij} is crucial to leveraging the exogenous variation provided by the centralized assignment mechanism. The first step to do so is to actually nail-down the assignment mechanism so that new draws of the lottery number yield counterfactual assignments that are consistent with the mechanism and its inputs.

8.1 Replicating School Offers in the Chilean System

I have made significant progress in being able to fully replicate the assignment delivered by the mechanism. On one hand this is possible by the fact that the responsible agency includes the used lottery number when making the data available. On the other hand, the problem becomes more complex given the number of features of the system that escape a traditional DA mechanism. Documentation for these is far from complete.

The Chilean system has many features that are not captured by the textbook DA mechanism which features ranked preferences and priorities $(\succ_i, \rho_i)$. There are reserved spots for low-income, high-achievement and special-needs students. Despite documentation not being clear about this, these quotas do roll-over to regular spots if they are not taken. Additionally, students have guaranteed admission in their original school if they do not get assigned anywhere else, even if their original school did not list any available spots. The system also allows siblings to apply in block, such that if the older one gets accepted by a school, the submitted list of younger siblings gets updated and prioritizes such school. Note this is different than the priority tier for students who have a sibling at the school before the process starts. Some of these rules also change over a transition period on the first few years of the implementation, which is where I sample from given that this allows me to have enough cohorts that have spent at least 4 years after their centralized assignment.

I can currently replicate 99.47 % of the assignments for almost 80k students applying to grade 9 in 2018. To get a sense of the importance of getting these additional features right, note that using only the the individual's types and a random seed replicates only up to 68 % of the assignments. Including the actual lottery numbers in this case, slightly decreases the fraction of matches.

Evident next steps in this part of the project include trying to figure out the remaining fraction, as well as ensure that I can achieve similar replication rates for later admission cycles, where some of the transition rules phase out.

8.2 Procedures to compute p_{ij}

The ideal way to compute p_{ij} in a pure DA mechanism is arguably to follow the mathematical derivation in **abdulkadiroglu_research_2017**. Given the importance of the additional features described in the previous section, however, I doubt this will be my preferred approach.

Many of the problematic features discussed above involve running variables, for which the framework in **abdulkadiroglu_breaking_2022** may be especially useful. However, some features—such as sibling block applications and guaranteed admission at the origin school—do not seem to fit naturally within that framework.

In the worst case, these values can still be computed through repeated simulations. At present, I am able to run more than 1,000 simulations of the DA algorithm for the Chilean setting in under an hour.